SIHYEON KIM

Master's Student in City and Regional Planning
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PROFESSIONAL SUMMARY

Urban planning and architecture researcher specializing in Urban Heat Island mitigation, pedestrian thermal comfort, and AI-aided urban climate planning. My research integrates microclimate simulation, geospatial analysis, machine learning, spatial optimization, and cost-effectiveness assessment to identify context-specific heat mitigation strategies. Experienced in leading first-author projects using ENVI-met, SOLWEIG, Python, GIS, and genetic algorithms to support climate-responsive urban planning and design.

EDUCATION

Seoul National University, Graduate School of Environmental Studies — Seoul, Korea
M.S. in City and Regional Planning — 2025 – Present

Soongsil University — Seoul, Korea
Bachelor of Architecture (B.Arch), 5-year program — 2025
Departmental Honors Award

PUBLICATIONS

Journal Articles and Manuscripts

Quan, S. J.*, & Kim, S. (2025). Mixture of urban form archetypes in Seoul: A signal processing approach. *Urban Design International*. **In press.**

Kim, S., & Quan, S. J.* (2026). Evaluation of cost-effectiveness of urban heat mitigation strategies in Seoul's urban regeneration projects. **First-author manuscript; full first draft completed and under revision for submission in 2026.**

Kim, S., & Quan, S. J.* (2026). Budget-constrained green-roof spatial optimization in three Seoul urban regeneration areas using SOLWEIG and genetic algorithms. **First-author research project initiated in September 2026; manuscript in preparation.**

** Corresponding author*

RESEARCH EXPERIENCE

Seoul National University — Graduate School of Environmental Studies — Seoul, Korea
Advisor: Professor Steven Jige Quan

Cost-Effective Urban Heat Mitigation across Seoul Urban Regeneration Typologies — Oct 2024 – Present
First-Author Researcher, City Energy Lab

- Led the full research workflow under the advisor's guidance on research questions and methodology, including statutory urban regeneration plan review, site selection, data collection, and study design.
- Built and cleaned spatial and tabular datasets using Python and GIS; applied PCA and k-means clustering to classify Seoul urban regeneration areas and select representative typologies.
- Developed and validated ENVI-met models to compare green roofs, bright roofs, and permeable pavement using pedestrian-level UTCI across daytime, evening, and full-day periods.
- Integrated installation and maintenance costs with thermal-performance results to evaluate cost-effectiveness and derive typology-specific planning recommendations.

Budget-Constrained Green-Roof Spatial Optimization Using SOLWEIG and Genetic Algorithms — Mar 2026 – Present
First-Author Researcher, City Energy Lab

- Reframed urban heat mitigation as a fixed-budget spatial allocation problem and led project design, data processing, simulation, optimization, results analysis, and policy interpretation.
- Prepared 2 m raster datasets and conducted unit-level SOLWEIG-GPU simulations for candidate rooftops across three Seoul urban regeneration areas.
- Constructed a precomputed ΔUTCI lookup table and implemented a binary genetic algorithm to maximize pedestrian cooling under six policy budget tiers.

- Evaluated meaningful cooling area, budget-performance curves, marginal returns, and saturation points to distinguish scalable, targeted, and low-sensitivity contexts.

Mixture of Urban Form Archetypes in Seoul: A Signal Processing Approach 2025 – Present

Second Author, City Energy Lab

- Led the qualitative component reviewing Seoul's development history and urban form typologies; synthesized approximately 60 studies across major historical periods.
- Connected SimplexNMF and clustering results with knowledge-based typologies through GIS overlays and a Sankey diagram linking mixture clusters to forms identified in the literature.
- Produced the historical timeline, diagrams, maps, illustrations, and visual materials supporting the qualitative–quantitative interpretation.

UN-Habitat–Collaborative Smart Inclusive Transition Project for Growing and Shrinking Cities Jul 2025

Research Assistant, International Collaborative Research Project

- Contributed to a UN-Habitat–collaborative project developing the Smart Inclusive Transition Index to compare smartness, inclusiveness, governance, digital innovation, and environmental sustainability across growing and shrinking East Asian cities.
- Collected, standardized, and cleaned multi-source urban data for six growing and shrinking cities in China and conducted Python-based comparative analyses to support city diagnosis and typology development.
- Assisted in translating city-level indicator results into comparative evidence for context-specific inclusive smart-city transition strategies.

Soongsil University — School of Architecture Design, College of Engineering Seoul, Korea

Advisor: Professor Soomi Kim

Design Method of Incineration Plant Architecture as Healing and Recovery Mar 2024

First Author; presented at the 2024 Architectural Institute of Korea Spring Conference, Undergraduate Session

- Reviewed adaptive-reuse cases of idle industrial facilities to derive healing- and recovery-oriented regeneration strategies for incinerator architecture.
- Developed a design framework positioning the incinerator as a mediating space where human, machine, and nature interact, and translated the research findings into the Mokdong Incinerator graduation design proposal.

TEACHING EXPERIENCE

Seoul National University — Graduate School of Environmental Studies Seoul, Korea

Advisor: Professor Steven Jige Quan

Teaching Assistant, Urban Form and Energy Spring 2026

- Delivered two hands-on simulation sessions: building energy consumption analysis with **DesignBuilder** and urban microclimate simulation with **ENVI-met.**
- Prepared instructional materials and guided students through model setup, simulation workflows, output interpretation, and troubleshooting.
- Managed attendance, assignment checks, and student inquiries throughout the course.

CONFERENCE PRESENTATIONS

Kim, S., & Quan, S. J.* (2026). Evaluation of Cost-Effectiveness of Urban Heat Mitigation Strategies in Seoul's Regeneration Projects. 66th ACSP Annual Conference, United States, October 10–12.

Kim, S., & Quan, S. J.* (2026). Cost-Effective Heat Mitigation across Urban Regeneration Typologies: Implications for Urban Microclimate in Seoul. 20th IACP Conference, Xi'an, China, July 10–12.

Kim, S., & Quan, S. J.* (2026). Thermal Performance and Cost-Effectiveness of Urban Heat Mitigation Strategies in Seoul's Regeneration Projects. 2nd Workshop on AI for Urban Planning, AAAI, Singapore, January 26.

Quan, S. J.*, & Kim, S. (2026). Revealing hidden archetypes of urban morphological patterns in Seoul. *2nd Workshop on AI for Urban Planning, AAAI*, Singapore, January 26. Presented by S. J. Quan.

Kim, S., & Quan, S. J.* (2025). Assessing Urban Heat Mitigation Strategies in Seoul's Regeneration Projects. 19th IACP Conference, Xiamen, China, July 3–7.

Kim, S., & Quan, S. J.* (2025). Evaluating the Effectiveness of Heat Mitigation Strategies in Seoul's Urban Regeneration Projects. UDIK Spring Conference, Chuncheon, Korea, pp. 792–795.

Kim, S., & Kim, S. M.* (2024). Design Method of Incineration Plant Architecture as Healing and Recovery. Architectural Institute of Korea Spring Conference, Undergraduate Session, March 13.

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HONORS & AWARDS

Academic & Research

- Best Cross-disciplinary Integration Award, Participatory Urban Regeneration Empowered by Artificial Intelligence — 2nd AI4CITIES International Conference & Summer Workshop, Tongji University (2025).

Architecture & Design

- Academic Excellence Award, Department of Architecture, Soongsil University (2025).
- Graduation Design Award, Soongsil University (2024).
- Encouragement Award, Modular Urban Regeneration Project, Korea Space Design Association National Competition (2023).
- Award Winner, Eco-friendly Office Building Design, Korean Society of Basic Design & Art; Sejong Center Exhibition (2023).

Fine Arts

- Three-Time Award Winner, JoongAng Painting Competition; selected works exhibited at the Korea Art Museum (2022–2024)
- Special Prize in Fine Art, 5th Art Korea World Art Competition (2026)
- Selected Exhibitor, 26th Korean Painting Status Exhibition (2026)
- Award Winner, Korean Women's Painting Competition (2026)
- Award Winner, Oil Painting Division, Korean Society of Basic Design & Art; exhibited at the Sejong Center (2023)
- First Prize, AP Art Awards; City Hall Winter Youth Art Exhibition, Tallahassee, Florida (2014–2015)

PROFESSIONAL EXPERIENCE

Design Somnium Seoul, Korea
Architecture Intern — Multi-purpose Cultural Center Remodeling Jan 2024 – Jul 2024

- Prepared daily construction reports and documented demolition, structural reinforcement, MEP, façade, and interior work.
- Supported design–construction coordination through CAD and SketchUp modeling and on-site communication.

SKILLS

Programming & Data Analysis: Python (Pandas, NumPy, GeoPandas, Scikit-learn, Matplotlib), statistical analysis, machine learning, genetic algorithms, optimization

GIS & Urban Analytics: ArcGIS Pro, QGIS, raster/vector processing, spatial databases, PCA, k-means clustering

Environmental Simulation & Optimization: ENVI-met, SOLWEIG, DesignBuilder, IES VE, micro climate analysis

Design & Visualization: AutoCAD, SketchUp, Rhino 3D, Grasshopper, Ladybug Tools, V-Ray, Adobe Photoshop, Illustrator, InDesign, Figma

CERTIFICATIONS & LANGUAGES

Rhino Certification (2025.08) | ESG SNU Youth Forum Certification (2026)

Korean: Native | English: Professional working proficiency

ARCHITECTURAL DESIGN PROJECTS

Soongsil University — School of Architecture, College of Engineering Seoul, Korea

Individual academic projects; independently completed concept development, site analysis, architectural design, drawings, diagrams, and final visualization.

Healing Fragmentation: Mokdong Incinerator Regeneration Graduation Project, 2024

Graduation Design Award , Soongsil University

- Reframed an essential but socially isolated waste facility as a mediating public landscape connecting people, nature, and industrial systems; developed a phenomenological healing sequence through changing viewpoints, movement, light, and adaptive reuse.

Urban Play Hub: Yongsan Electronics Market Regeneration Urban Design Studio, 2023

- Used AI-assisted massing exploration to regenerate a declining electronics district around productive play, healing, logistics, and community programs; reorganized vehicle, pedestrian, and freight circulation at the master-plan scale.

Circulation of Communication: Mobility-Based Yongsan Renewal Competition Project, 2023

Encouragement Award, Korea Space Design Association National Competition

- Proposed PBV mobility units, modular mixed-use additions, a pedestrian linear plaza, and separated underground logistics to demonstrate future-oriented urban design responsive to changing retail, work, and housing patterns.

Smart Farm + Skin: Eco-Friendly Office Environmental Design Studio, 2022

Award Winner, Korean Society of Basic Design & Art

- Integrated smart farms, terraces, and a double-skin facade; used IES VE to compare perforation, louver, and window-size alternatives for daylight, natural ventilation, glare control, and reduced lighting-energy demand.

Connecting Multi Paths into Residents' Heart Housing Studio, 2020

Outstanding Architectural Design Award , Soongsil University

- Designed a stepped artist-housing community in Huam-dong that preserved local topography and urban scale while reconnecting fragmented green space through shared courtyards, private gardens, ateliers, and extended alleys.

Soyo-yu (逸遙遊): Museum Remodeling and Structural Reuse Adaptive Reuse Studio, 2021

- Preserved the primary core and columns while creating a vertical void, ramps, sunken spaces, and a sequence of metal, wood, and light galleries to reinterpret the work and cultural identity of architect Itami Jun.

ADDITIONAL EXPERIENCE

- Sign Language Interpreter Volunteer — Real-time interpretation for church services (2019–Present).
- Volunteer Mentor, Cambodia — Educational program for elementary school students (August 2019).
- International Education: Chiles High School and Deer Lake Middle School, Florida; Meeker Elementary School, Iowa.